CS341 · COMPUTER GRAPHICS

# Mock Midterm Practice

20 marks · 1 hour · same format as the past midterms

**Muharib**

## 01 Questions

**Q1** SHORT ANSWER · 4 PTS

1. What is a frame buffer?
2. Define persistence. Which kind of persistence is better for animation?
3. P = (4, 1) and Q = (1, 5). Find the vector from P to Q and its length.
4. A 10 × 6 inch screen has a resolution of 1280 × 720. Find the aspect ratio.

**Q2** TRUE / FALSE · 4 PTS

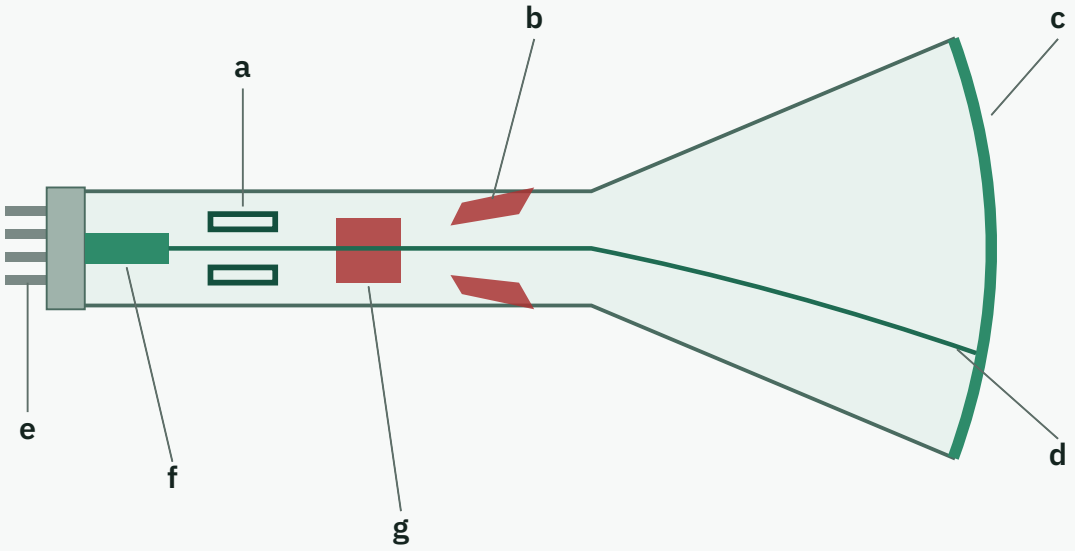
1. In a random scan system the electron beam directly draws the picture. ( )
2. A plasma panel is a nonemissive display. ( )
3. Point – Point = Vector. ( )
4. In a raster system, picture information is stored in the frame buffer. ( )
5. Passive-matrix displays are cheaper than active-matrix displays. ( )
6. Bresenham's algorithm is more accurate than DDA because it avoids rounding. ( )
7. Persistence is the time for emitted light to decay to half of its intensity. ( )
8. The dot product of two vectors is a vector. ( )

**Q3** MCQ · 4 PTS

1. Which device is a hard-copy device? a) Trackball b) Ink-jet printer c) Light pen d) Data glove
2. Which device draws lines directly with the electron beam? a) LCD b) Plasma c) Random scan d) Raster scan
3. Dot pitch is: a) The number of pixels b) The space between centers of adjacent pixels c) The refresh frequency d) The beam voltage
4. Two vectors are perpendicular when their dot product is: a) 1 b) –1 c) 0 d) Their length
5. A point plus a vector is: a) A vector b) A point c) A scalar d) Not defined
6. glVertex2i(4, 5) specifies: a) 2 floats b) 2 integers c) 3 integers d) An integer array
7. Lines in raster systems look stair-stepped because: a) Low refresh rate b) Positions are rounded to pixels c) Phosphor decay d) Color mixing
8. Plasma panels are: a) Nonemissive displays b) Emissive displays c) Hard-copy devices d) Input devices

**Q4** CRT · 3 PTS

Label the cathode ray tube using the letters a–g:



**Q5** BRESENHAM · 5 PTS

Using Bresenham's line algorithm, generate the pixels of the line from (1, 1) to (8, 5) and draw it. Show the steps.

| k | $p_k$ | Decision | Next pixel | $p_{k+1}$ |
|---|---|---|---|---|
| 0 | | | | |
| 1 | | | | |
| 2 | | | | |
| 3 | | | | |
| 4 | | | | |
| 5 | | | | |
| 6 | | | | |

